

EXP 6 >  create_excel.py > ...

```
1  import openpyxl
2
3
4  def generate_sample_data(file_path="students.xlsx"):
5      wb = openpyxl.Workbook()
6      ws = wb.active
7      ws.title = "StudentRecords"
8
9      # Header Row
10     ws.append(["Roll No", "Name", "Java", "Python", "DBMS"])
11
12     # Sample Dataset from Experiment Sheet
13     data = [
14         [101, "Amit", 75, 68, 80],
15         [102, "Neha", 58, 62, 55],
16         [103, "Raj", 45, 52, 61],
17         [104, "Sneha", 85, 72, 91],
18         [105, "Ajay", 59, 57, 60],
19         [106, "Pooja", 65, 48, 55],
20         [107, "Rohan", 61, 66, 64],
21         [108, "Anita", 39, 45, 50],
22         [109, "Kiran", 70, 75, 80],
23         [110, "Priya", 60, 59, 58]
24     ]
25
26     for row in data:
27         ws.append(row)
28
29     wb.save(file_path)
30     print(f"File '{file_path}' created successfully!")
31
32
33 if __name__ == "__main__":
34     generate_sample_data()
35
```

```
1  import openpyxl
2
3  def analyze_student_marks(file_path):
4      """
5      Reads student records from Excel and returns counts of students scoring > 60:
6      - in at least one subject
7      - in all subjects
8      """
9      wb = openpyxl.load_workbook(file_path)
10     sheet = wb.active
11
12     count_any_subject = 0
13     count_all_subjects = 0
14
15     # Iterate through data rows (skipping header at row 1)
16     for row in sheet.iter_rows(min_row=2, values_only=True):
17         if not row or row[0] is None:
18             continue
19
20         # Extract marks for Java, Python, DBMS
21         marks = [row[2], row[3], row[4]]
22
23         # Condition 1: Scored > 60 in AT LEAST ONE subject
24         if any(mark > 60 for mark in marks):
25             count_any_subject += 1
26
27         # Condition 2: Scored > 60 in ALL subjects
28         if all(mark > 60 for mark in marks):
29             count_all_subjects += 1
30
31     wb.close()
32     return count_any_subject, count_all_subjects
33
34
35 if __name__ == "__main__":
36     any_count, all_count = analyze_student_marks("students.xlsx")
37     print(f"Students with > 60 in ANY one subject: {any_count}")
38     print(f"Students with > 60 in ALL subjects: {all_count}")
39
```

EXP 6 > test_student_filter.py > TestStudentFilter > tearDownClass

```
1 import os
2 import unittest
3
4 from create_excel import generate_sample_data
5 from student_filter import analyze_student_marks
6
7
8 class TestStudentFilter(unittest.TestCase):
9     TEST_FILE = "test_students.xlsx"
10
11     @classmethod
12     def setUpClass(cls):
13         # Setup test spreadsheet prior to testing
14         generate_sample_data(cls.TEST_FILE)
15
16     @classmethod
17     def tearDownClass(cls):
18         # Clean up created file after testing
19         if os.path.exists(cls.TEST_FILE):
20             os.remove(cls.TEST_FILE)
21
22     def test_sample_data_counts(self):
23         """Black-Box Test: Verify counts match calculated expected values."""
24         any_count, all_count = analyze_student_marks(self.TEST_FILE)
25
26         # Expected from sample data:
27         # Any > 60: Amit, Neha, Raj, Sneha, Pooja, Rohan, Kiran = 7
28         # All > 60: Amit, Sneha, Rohan, Kiran = 4
29         self.assertEqual(any_count, 7, "Count for ANY subject > 60 is incorrect")
30         self.assertEqual(all_count, 4, "Count for ALL subjects > 60 is incorrect")
31
32     def test_file_not_found_exception(self):
33         """White-Box Test: Verify exception handling when file does not exist."""
34         with self.assertRaises(FileNotFoundError):
35             analyze_student_marks("non_existent_file.xlsx")
36
37
38 if __name__ == "__main__":
39     unittest.main()
```

- PS C:\Users\Afroz\Desktop\New folder\EXP 6> py test_student_filter.py
File 'test_students.xlsx' created successfully!

..

Ran 2 tests in 0.036s

OK

- PS C:\Users\Afroz\Desktop\New folder\EXP 6> █